
Linear Probes on Curved Latent Spaces

Anonymous Author(s)

Affiliation

Address

email

Abstract

Linear probes are commonly used to evaluate learned representations and infer physical quantities for objects without direct measurements. However, their global predictive performance does not reveal where individual predictions are locally reliable or how the geometry of the learned representation contributes to their errors. Here we identify a geometric source of local readout error. Although a linear probe is affine in the ambient space, its restriction to a curved embedding manifold has a nonzero intrinsic Hessian given by $\langle w_N, \Pi \rangle$, the manifold’s bending contracted with the probe’s normal component. We turn this classical identity into a per-object measurement on galaxy embeddings from five encoders, using a decoder-based estimate of the second fundamental form validated on a synthetic surface of known curvature. Comparing this probe-facing curvature with the intrinsic Hessian of the physical target, we find that their mismatch predicts local readout accuracy for magnitude and photometric redshift, robustly under cross-fitting and across encoders. A counterfactual intervention on a local second-order surrogate further shows that reversing the probe-facing bending reduces local accuracy at nearly every anchor, whereas randomly oriented bending of matched strength does not exhibit the same asymmetry. The resulting tensor mismatch provides a geometric diagnostic of local linear-readout reliability and helps identify which physical quantities exhibit geometry-associated variation in prediction accuracy.

1 Introduction

Linear probes evaluate learned representations by measuring how accurately targets can be recovered from frozen embeddings [Alain and Bengio, 2016]. Their global performance, however, does not reveal where individual predictions are reliable or how representation geometry contributes to their errors. This matters in physical science, where probes infer quantities without direct measurements. For galaxy images, linear probes predict catalogue quantities such as magnitude, redshift and stellar mass [Smith et al., 2024], but offer no local reliability diagnostic. We show that an ambient affine probe inherits curvature from the embedding manifold in its normal direction. The mismatch between this probe-facing curvature and the target’s intrinsic Hessian captures a local source of prediction error.

We contribute (i) a decoder-based pointwise estimate of the second fundamental form, validated on a synthetic surface of known curvature; (ii) the application of the classical identity $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle$ to a fitted probe, with a residual expansion identifying its second-order approximation error; and (iii) evidence from 86,471 galaxies and five encoders that curvature mismatch predicts local readout accuracy for magnitude and redshift, supported by an intervention on a local second-order surrogate with a matched random-direction control.

2 Pointwise curvature from a decoder

Data. 86,471 galaxies from the AstroPT galaxy set [Smith et al., 2024] with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass), embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below) and by four further encoders (Section 4); embeddings from the Platonic Universe release [Duraphe et al., 2025]. We unit-normalize embeddings (data on S^{D-1} ; $D = 768$ for ViT-B).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^D$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, all by automatic differentiation (cf. [Lee and Park, 2023, Acosta et al., 2023]) at points where DF has full column rank d . We differentiate the projected decoder $F/\|F\|$, whose image lies on the unit sphere, so that Π splits exactly into the sphere’s universal radial part and an in-sphere shape part (Eq. (2); Appendix E).

Validation. On a synthetic in-sphere surface at $D = 768$ with closed-form Π (Appendix E), the decoder’s estimate matches the true trace of Π in direction (cosine 0.999), magnitude (ratio 1.00) and rank across points (0.94).

3 What a linear readout sees on a curved manifold

Notation. Let $M \subset \mathbb{R}^D$ be a d -dimensional smooth embedded manifold, $d < D$, with anchor $x_0 \in M$. Let J be an orthonormal tangent frame at x_0 and u normal coordinates in this frame, so that locally $x(u) = x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(\|u\|^3)$ [Lee, 2018], where Π is the normal-valued second fundamental form. In the numerical estimates, $J = DF$ is the decoder Jacobian, with metric $g = J^\top J$, and tangent-projected coordinates are used. A probe weight decomposes as $w = w_T + w_N$ into tangent and normal components. On the unit sphere, $\Pi = \Pi^S - gx$ separates the in-sphere shape from the universal radial bending, and w_S is the component of w_N orthogonal to x . Let y be a smooth target on M and $\hat{y} = w \cdot x + b_0$ a global affine probe. Tensor norms and inner products use $\langle A, B \rangle_g = g^{ia}g^{jb}A_{ij}B_{ab}$.

Three facts follow. First, although the probe is affine in the ambient space, its restriction to M is generally nonlinear in intrinsic coordinates:

$$(d\hat{y})_J = J^\top w, \quad \text{Hess}_M \hat{y} = \langle w_N, \Pi \rangle =: K. \quad (1)$$

We call K the probe-facing curvature. Second, on the unit sphere it splits into an in-sphere shape term and a radial term:

$$K = \underbrace{\langle w_S, \Pi^S \rangle}_{K_S} + \underbrace{[-(\hat{y} - b_0)g]}_{K_{\text{sph}}}. \quad (2)$$

Thus, even an ambient affine readout inherits intrinsic curvature from the geometry of the representation.

Third, consider an isotropic Gaussian patch $u \sim N(0, s^2 I_d)$ in the orthonormal tangent frame. For sufficiently smooth embedding and target functions, with controlled higher derivatives, let $\bar{c} = \mathbb{E}_{\text{patch}}[y - \hat{y}]$ be the patch-mean residual, $\delta = (dy - d\hat{y})_J$ the differential mismatch, and $\Delta = \text{Hess}_M y - K$ the Hessian mismatch. Then

$$\begin{aligned} \mathbb{E}_{\text{patch}}[(y - \hat{y})^2] &= \bar{c}^2 + s^2 \|\delta\|^2 \\ &\quad + \frac{1}{2}s^4 \|\Delta\|_g^2 + O(s^4 \|\delta\|) + O(s^6). \end{aligned} \quad (3)$$

The Hessian mismatch therefore determines the quadratic residual’s contribution to centered local error under this model. Its contribution is smaller than at $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle_g > \|K\|_g^2$: the probe-facing curvature must bend toward the target’s curvature, without overshooting it. With the sphere term fixed, the corresponding condition for the shape term is $2\langle R, K_S \rangle_g > \|K_S\|_g^2$, where $R = \text{Hess}_M y - K_{\text{sph}}$. Section 4 tests this relationship. Since the quadratic contribution scales as s^4 , our empirical analysis controls for variation in local neighborhood scale.

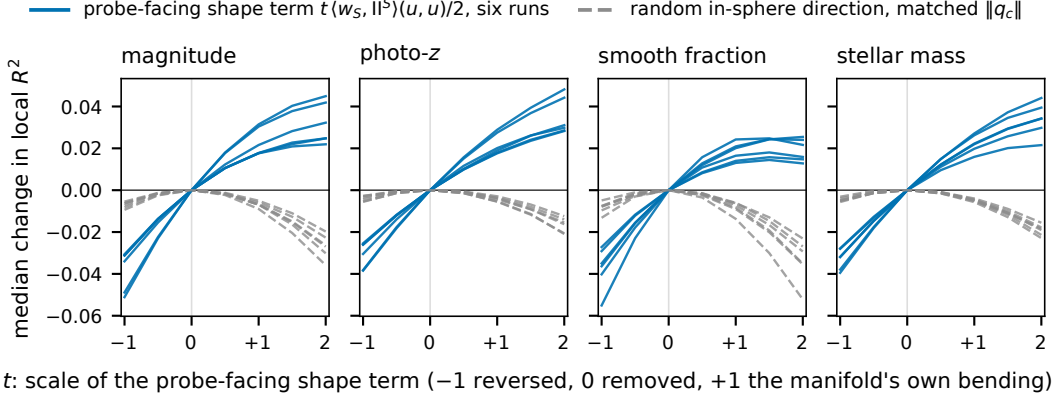


Figure 1: Bending toward the label helps, bending away hurts. At each of 512 anchors the readout’s in-sphere shape quadratic $\frac{t}{2} \langle w_S, \Pi^S \rangle(u, u)$ is scaled by t in a local second-order surrogate at a fixed manifold; lines are the median over anchors of the change in local R^2 relative to shape-flat ($t = 0$). $t = 1$ is the manifold’s own bending seen from the readout’s normal direction, $t = -1$ its reversal. Blue: the decoder’s shape term, one line per run (five encoders at $d = 16$, ViT-B also at $d = 20$). Grey dashed: a random in-sphere normal direction with matched quadratic amplitude $\|q_c\|$. Fractions of anchors helped and hurt, and sign tests on non-overlapping anchors, in Appendix D.

4 Probe-facing curvature: bending toward the label helps, bending away hurts

Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 with Freedman–Lane p -values [Freedman and Lane, 1983], controlling for log patch radius, local label variance and count and, stronger, log radius at five scales.

Galaxies. We estimate $\text{Hess}_M y$ at each anchor by fitting a least-squares quadratic to the labels of its 2,048 neighbours in tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$. These decoder-based Monge coordinates have vanishing Christoffel symbols at the anchor, so the quadratic coefficient estimates the covariant Hessian in the small-neighbourhood limit. For magnitude and redshift, the mismatch $\|\Delta\|_g$ is negatively associated with local R^2 (-0.39 to -0.45), while alignment $\cos_g(\text{Hess}_M y, K)$ is positively associated with it ($+0.24$ to $+0.43$). These relationships persist under cross-fitting (Appendix B). Smooth fraction exhibits weaker associations, while stellar mass shows no significant mismatch or alignment signal in this experiment (Table 3; decoder ablations, Appendix A). Across five encoders, mismatch remains negatively associated with local accuracy for magnitude and redshift, while alignment is positive and significant for magnitude across all five and redshift across four (Appendix C).

The shape term closely tracks the magnitude of the total probe-facing curvature on galaxies (rank correlation 0.96–0.99), but its association with local accuracy varies across targets. This is consistent with Eq. (3), where the shape term’s contribution to the quadratic mismatch depends on both its magnitude and its alignment with the remaining target curvature. The sphere term is negatively associated with local accuracy for all four targets in the main ViT-B experiment, although its association is sensitive to probe regularization for some targets (Appendix E). Applying the same local quadratic regression to the probe predictions provides a decoder-independent estimate of K , with median tensor cosine 0.55–0.65 relative to the decoder-based estimate.

Relation to prior work. The identity in Eq. (1) is classical [Milnor, 1963, Chern and Lashof, 1957]; related work derives a curvature-dependent normal weight for least-squares

111 regression on hypersurfaces [Liu et al., 2023] and documents the effects of curved repre-
 112 sentations on linear decoding [Jurewicz et al., 2024]. Other approaches study manifold
 113 capacity [Chung et al., 2016, Slatton et al., 2026], representation flattening [Psenka et al.,
 114 2024], curvature regularization [Ma et al., 2023], latent geometry [Kaufman and Azencot,
 115 2023], local-regression bias [Cheng and Wu, 2013], and geometric indicators of generaliza-
 116 tion [Chou et al., 2026]. Our contribution is to estimate the pointwise probe-facing curvature
 117 tensor on a learned manifold, compare it with the target’s intrinsic Hessian, and test its
 118 contribution to local readout accuracy through a controlled second-order intervention.

119 **Bending toward helps, bending away hurts.** To isolate the contribution of probe-
 120 facing shape curvature, we intervene on the in-sphere quadratic term of a local second-order
 121 surrogate while keeping the manifold, tangent readout, and sphere term fixed. Each anchor’s
 122 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2}K_{\text{sph}}(u, u) + \frac{t}{2}K_S(u, u)$, where a is the
 123 probe’s tangent differential, K_S is estimated from the decoder, and c is refit for each t . Here
 124 $t = 1$ retains the estimated shape quadratic of the original readout, $t = -1$ reverses its sign,
 125 and $t = 0$ removes it.

126 Under the isotropic Gaussian quadratic model, $t = 1$ improves upon $t = 0$ iff $2\langle R, K_S \rangle_g >$
 127 $\|K_S\|_g^2$. On the actual neighbours, let r_0 be the centred shape-flat residual and q_c the centred
 128 quadratic contribution $\frac{1}{2}K_S(u_i, u_i)$. After refitting the intercept, the exact finite-sample
 129 criterion is $2\langle r_0, q_c \rangle > \|q_c\|^2$, with optimum $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ (Appendix D).

130 Across five encoders and all four labels (Figure 1), $t = 1$ improves local R^2 relative to the
 131 shape-flat surrogate at 79–99% of anchors, while $t = -1$ reduces it at 97–100%. A random
 132 in-sphere normal direction with matched centred quadratic amplitude shows no comparable
 133 directional asymmetry and is typically harmful under either sign ($t = 1$ improves upon
 134 shape-flat at 22–36% of anchors; Appendix D).

135 Thus, the probe-selected shape quadratic is locally error-reducing within the second-order
 136 surrogate. Under the quadratic Gaussian model, this is the signature of bending toward
 137 the target’s remaining intrinsic curvature. This within-anchor comparison differs from the
 138 cross-anchor associations in Table 3 and can hold even where the latter show no significant
 139 signal, as for stellar mass. The optimum t^* lies between 1.5 and 3.6 across runs, indicating
 140 that further amplification of the shape term reduces local surrogate error.

141 5 Discussion

142 Our results identify a geometric source of local linear-probe error: the mismatch $\Delta =$
 143 $\text{Hess}_M y - \langle w_N, \Pi \rangle$ between the target’s intrinsic Hessian and the curvature induced by the
 144 probe. We estimate this mismatch using a decoder-based second fundamental form, validated
 145 on a synthetic surface and showing moderate agreement with a decoder-independent estimate
 146 on galaxy embeddings. Across five encoders, mismatch tracks local accuracy for magnitude
 147 and redshift, with weaker evidence for smooth fraction. A counterfactual intervention on
 148 the local second-order surrogate shows that the probe-selected shape term reduces error at
 149 most anchors, while reversing it generally increases error. Thus, manifold curvature can help
 150 rather than hinder linear decoding when it bends toward the target’s curvature.

151 The mismatch can be estimated from labelled neighbours without knowing an object’s target
 152 value, providing a local diagnostic of predictions that may warrant further scrutiny. It is
 153 not yet a calibrated per-object uncertainty estimate. For stellar mass, mismatch does not
 154 consistently track local accuracy despite the beneficial shape contribution, showing that a
 155 locally useful geometric effect need not explain variation in overall readout accuracy.

156 *Limitations.* We study one galaxy sample, four targets and five encoders. The label Hessian
 157 is estimated from finite neighbourhoods; its norm is reliably ranked (split-half $\rho = 0.91$ – 0.94),
 158 but its direction is only moderately recovered (tensor cosine 0.19 – 0.35). The synthetic
 159 surface does not independently corroborate the alignment association because its target
 160 has insufficient intrinsic curvature. The intervention modifies a local surrogate, not the
 161 encoder or manifold. Mismatch and alignment are the primary hypotheses; other analyses
 162 are exploratory. Overlapping neighbourhoods introduce anchor dependence that may affect
 163 permutation-test validity.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. arXiv:1610.01644.
- Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large observation models for astronomy. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2405.14930. URL <https://arxiv.org/abs/2405.14930>.
- Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025. URL <https://arxiv.org/abs/2509.19453>.
- Yonghyeon Lee and Frank C. Park. On explicit curvature regularization in deep generative models. In *Proceedings of the 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, volume 221 of *Proceedings of Machine Learning Research*, pages 505–518. PMLR, 2023.
- Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic curvature in neural manifolds. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 610–619, 2023. doi: 10.1109/CVPRW59228.2023.00068.
- John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer, 2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.
- David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.
- John Milnor. *Morse Theory*, volume 51 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, 1963. doi: 10.1515/9781400881802. Based on lecture notes by M. Spivak and R. Wells.
- Shiing-Shen Chern and Richard K. Lashof. On the total curvature of immersed manifolds. *American Journal of Mathematics*, 79(2):306–318, 1957. doi: 10.2307/2372684.
- Liangchen Liu, Juncai He, and Yen-Hsi Tsai. Linear regression on manifold structured data: the impact of extrinsic geometry on solutions. In *Proceedings of the 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, volume 221 of *Proceedings of Machine Learning Research*, pages 557–576. PMLR, 2023.
- Katarzyna Jurewicz, Brianna J. Sleezer, Priyanka S. Mehta, Benjamin Y. Hayden, and R. Becket Ebitz. Irrational choices via a curvilinear representational geometry for value. *Nature Communications*, 15(1):6424, 2024. doi: 10.1038/s41467-024-49568-4.
- SueYeon Chung, Daniel D. Lee, and Haim Sompolinsky. Linear readout of object manifolds. *Physical Review E*, 93(6):060301, 2016. doi: 10.1103/PhysRevE.93.060301.
- Will Slatton, Chi-Ning Chou, and SueYeon Chung. Linear readout of neural manifolds with continuous variables, 2026. URL <https://arxiv.org/abs/2603.10956>.
- Michael Psenka, Druv Pai, Vishal Raman, Shankar Sastry, and Yi Ma. Representation learning via manifold flattening and reconstruction. *Journal of Machine Learning Research*, 25(132):1–47, 2024. URL <https://www.jmlr.org/papers/v25/23-0615.html>.
- Yanbiao Ma, Licheng Jiao, Fang Liu, Shuyuan Yang, Xu Liu, and Lingling Li. Curvature-balanced feature manifold learning for long-tailed classification. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15824–15835, 2023. doi: 10.1109/CVPR52729.2023.01519.
- Ilya Kaufman and Omri Azencot. Data representations’ study of latent image manifolds. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 15928–15945. PMLR, 2023.
- Ming-Yen Cheng and Hau-Tieng Wu. Local linear regression on manifolds and its geometric interpretation. *Journal of the American Statistical Association*, 108(504):1421–1434, 2013. doi: 10.1080/01621459.2013.827984.
- Chi-Ning Chou, Artem Kirsanov, Yao-Yuan Yang, and SueYeon Chung. Diagnosing generalization failures from representational geometry markers. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://arxiv.org/abs/2603.01879>.

218 Appendix A: decoder ablations

219 Table 1 repeats the four quantities of Tables 3 and 7 at $d = 16$ under the multi-scale density
 220 control for two further decoder initialisation seeds and for a decoder of width 400^3 instead
 221 of 250^3 (all fits reach variance explained 0.952). Every significant sign of the main table is
 retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	-0.26	-0.26	-0.27	-0.26
	mismatch	-0.39	-0.35	-0.34	-0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	-0.06*	-0.09*	-0.09	-0.06*
	sphere	-0.33	-0.35	-0.34	-0.35
	mismatch	-0.45	-0.43	-0.42	-0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	-0.13	-0.15	-0.14	-0.11
	mismatch	-0.09*	-0.05*	-0.08*	-0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	-0.06*	+0.05*	+0.09
	sphere	-0.35	-0.35	-0.35	-0.35
	mismatch	-0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	-0.02*	+0.03*	-0.02*

Table 1: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

222

223 Appendix B: sensitivity of the mismatch and sphere columns

224 Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half
 225 and local R^2 scored on the other (both directions shown), so the Hessian estimate and the
 226 outcome share no rows. The split-half cosine between the two Hessian estimates is the
 227 reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global
 228 out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme
 predictions; the sphere term is recomputed from that probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 2: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

229

230 Appendix C: Main and cross-encoder results

231 **Main results.** Table 3 reports the associations between geometric mismatch, alignment
 232 and local readout accuracy for ViT-B at $d = 16$ and $d = 20$. Results use 512 anchors,
 233 $k = 2,048$ neighbours and the multi-scale density control.

label	mismatch $\ \Delta\ _g$		alignment $\cos_g(\text{Hess}_M y, K)$	
	$d=16$	$d=20$	$d=16$	$d=20$
mag_r	−0.39	−0.40	+0.35	+0.43
photo_z	−0.45	−0.44	+0.26	+0.24
smooth_fraction	−0.09*	−0.17	+0.19	+0.07*
stellar_mass	−0.04*	−0.09*	+0.01*	+0.02*

Table 3: ViT-B: rank-partial Spearman correlations with local R^2 under the multi-scale density control. * Not significant at 0.05. Shape, sphere and mean-curvature results are reported in Appendix E.

Across encoders. We repeat the geometric diagnostic at $d = 16$ on the same 86,471 galaxies using four further encoders from the Platonic Universe release: DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B and ViT-L (widths 768, 512, 1,024 and 1,024; all unit-normalized). A decoder of the same architecture and training protocol is fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965 and 0.968, respectively). All analyses use 512 anchors, $k = 2,048$ and the multi-scale density control.

For magnitude and redshift, mismatch is negatively associated with local R^2 across all five encoders (−0.32 to −0.61; cross-fitted −0.28 to −0.62). Alignment is positive and significant for magnitude across all five (+0.13 to +0.53) and for redshift across four. Shape-term associations vary by encoder and target, while smooth fraction shows less consistent mismatch and alignment signals. For stellar mass, alignment is nonsignificant across encoders and mismatch significant in only one main comparison. Thus, the cross-anchor mismatch association varies by target, even where the shape term improves the local surrogate.

Table 4 reports the full cross-encoder results, including global out-of-sample probe R^2 in the first row for each label. Table 5 reports cross-fitted mismatch and alignment associations and split-half reliability of the estimated label Hessian.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	−0.11	+0.16	+0.16	−0.21
	sphere	−0.26	−0.23	−0.40	−0.24	−0.08*
	mismatch	−0.39	−0.57	−0.34	−0.32	−0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	−0.06*	−0.05*	−0.17	+0.29	−0.16
	sphere	−0.33	−0.21	−0.29	−0.46	−0.44
	mismatch	−0.45	−0.61	−0.45	−0.48	−0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	−0.08*	+0.20	+0.24	+0.06*
	sphere	−0.13	−0.06*	+0.05*	−0.15	−0.11
	mismatch	−0.09*	−0.33	−0.25	−0.20	−0.01*
	alignment	+0.19	−0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	−0.15	−0.10	−0.11	−0.13
	sphere	−0.35	−0.26	−0.38	−0.40	−0.38
	mismatch	−0.04*	+0.03*	−0.12	−0.06*	−0.02*
	alignment	+0.01*	−0.05*	−0.01*	+0.07*	−0.03*

Table 4: Cross-encoder results at $d = 16$: global probe R^2 and rank-partial Spearman correlations with local R^2 under the multi-scale density control. * Not significant at 0.05.

Appendix D: intervening on the readout’s normal component

The partials of Table 3 compare anchors with one another and cannot say whether reversing the probe-facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual local second-order surrogate of the readout at a fixed manifold. At

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	-0.60	-0.62	+0.50	+0.50	+0.35
	photo_z	-0.56	-0.59	+0.28	+0.27	+0.24
	smooth_fraction	-0.32	-0.29	-0.03*	-0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	-0.03*	-0.07*	+0.22
CLIP-B	mag_r	-0.28	-0.35	+0.28	+0.29	+0.31
	photo_z	-0.41	-0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	-0.22	-0.24	+0.15	+0.20	+0.28
	stellar_mass	-0.14	-0.13	-0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	-0.36	-0.34	+0.19	+0.21	+0.29
	photo_z	-0.46	-0.45	+0.18	+0.19	+0.30
	smooth_fraction	-0.15	-0.19	-0.02*	+0.03*	+0.32
	stellar_mass	-0.05*	-0.01*	+0.14	-0.03*	+0.29
ViT-L	mag_r	-0.46	-0.49	+0.09*	+0.15	+0.30
	photo_z	-0.44	-0.42	+0.09*	+0.13	+0.24
	smooth_fraction	-0.03*	-0.04*	+0.11	+0.07*	+0.30
	stellar_mass	-0.06*	+0.01*	-0.04*	-0.03*	+0.21

Table 5: Cross-fitted mismatch and alignment partials under the multi-scale density control, with median split-half tensor cosine for the estimated label Hessian. * Not significant at 0.05.

each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; we score the anchor’s 2,048 neighbours by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the local intercept refit, where $q = \frac{1}{2}\langle w_S, \Pi^S \rangle(u, u)$ is the decoder’s in-sphere second-order term in the anchor’s chart coordinates: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction, $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). Because the intercept is refit for every t , the local sum of squares is exactly $\text{SSE}(t) = \|r_0 - tq_c\|^2$ with r_0 the centred residual of the shape-flat predictor and $q_c = q - \bar{q}\mathbf{1}$ the centred quadratic on the anchor’s neighbours; hence $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ and $t = 1$ beats shape-flat iff $2\langle r_0, q_c \rangle > \|q_c\|^2$. This is the finite-sample counterpart of the tensor-level condition $2\langle R, K_S \rangle_g > \|K_S\|_g^2$ of Section 3, not the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not. Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than shape-flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random in-sphere normal direction v whose quadratic is rescaled to the same centred amplitude on the actual neighbours, $\|q_{v,c}\| = \|q_c\|$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; matching $\|\langle v, \Pi^S \rangle\|_g$ gives the same picture; Supplement 12). The anchors’ neighbourhoods overlap (each point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal set of anchors whose pairwise overlap is at most 5% of k has 19–21 members per run, on which $t = 1$ beats shape-flat at 81–100% (one-sided sign test $p \leq 4 \times 10^{-3}$ in every cell) and $t = -1$ is worse at 95–100% ($p \leq 3 \times 10^{-5}$).

Appendix E: the shape and sphere terms, and mean curvature

The mean-curvature vector of the decoder surface is $H = \text{tr}_g \Pi$ (unnormalized trace; some authors divide by d). Alongside Eq. (1), $\Delta_M(wx) = \langle w, H \rangle$: the Laplacian of a linear readout sees only the trace of Π , its Hessian the whole tensor, so $\|H\|$ need not track $\langle w_N, \Pi \rangle$. On the unit sphere the Euclidean mean-curvature vector of every d -submanifold has the universal radial component $-d\hat{x}$, which says nothing about shape; differentiating the projected decoder $F/\|F\|$ splits $H = -d\hat{x} + H^S$ exactly (radial part $-d$ to 10^{-14}), and H^S is the non-radial mean-curvature component used below.

Validation surface. At $D = 768$ the in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0\dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a rotation) has closed-form curvature; the decoder’s H^S gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999, magnitude ratio 1.00. Resampling this surface

run	label	decoder in-sphere shape term					random orientation, matched $\ q_c\ $			
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.26	0.80	-0.009	-0.009
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.33	0.68	-0.005	-0.006
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.78	-0.008	-0.010
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.29	0.70	-0.006	-0.006
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.24	0.78	-0.007	-0.008
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.36	0.71	-0.004	-0.005
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.22	0.77	-0.007	-0.008
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.32	0.72	-0.005	-0.005
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.29	0.71	-0.007	-0.006
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.29	0.71	-0.006	-0.004
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.22	0.78	-0.014	-0.013
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.26	0.74	-0.005	-0.005
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.30	0.68	-0.006	-0.005
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.36	0.64	-0.003	-0.003
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.25	0.69	-0.006	-0.005
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.35	0.68	-0.004	-0.005
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.26	0.73	-0.007	-0.007
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.28	0.64	-0.004	-0.003
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.23	0.75	-0.009	-0.008
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.29	0.65	-0.005	-0.003
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.74	-0.005	-0.006
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.34	0.67	-0.004	-0.003
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.25	0.77	-0.006	-0.008
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.30	0.67	-0.004	-0.004

Table 6: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at a large majority of anchors, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.

with weight $\|H^S\|^\gamma$ (Figure 2a) moves the density-controlled $\|H^S\|$ partial against local R^2 from -0.29 to $+0.26$, while $\|\langle w_N, \Pi \rangle\|_g$ holds at -0.74 to -0.80 , dominated by the sphere term of Eq. (2); the mismatch partial reproduces there, the alignment partial has no stable sign.

Galaxies: shape and sphere terms. Table 7 gives the density-controlled partials of the two terms of Eq. (2) against local R^2 (Figure 2b). The shape term dominates the probe-facing Hessian on galaxies (rank 0.96–0.99 with $\|K\|_g$) and is label-signed: positive for magnitude and smooth fraction, null to weakly negative for redshift, null for stellar mass. The sphere term is negative on every label and both d (-0.13 to -0.36) under the $\alpha = 100$ probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass, so part of it is ridge shrinkage of extreme predictions rather than geometry.

Galaxies: mean curvature. The $\|H^S\|$ partial changes sign with the latent dimension for magnitude and is null for stellar mass. Across the five encoders of Appendix C the quadratic-chart mean-bending trace has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. Mean curvature, the scalar usually reported, keeps only the trace of Π and fails accordingly: its association with readout accuracy changes sign with the sampling on the known surface and with the latent dimension and the encoder on galaxies.

label	shape $\ \langle w_S, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		$\ H^S\ $	
	$d=16$	$d=20$	16	20	16	20
mag_r	+0.12	+0.19	-0.26	-0.26	+0.25	-0.17
photo_z	-0.06*	-0.12	-0.33	-0.36	+0.28	+0.13
smooth_fraction	+0.07*	+0.16	-0.13	-0.15	+0.21	+0.15
stellar_mass	+0.00*	+0.03*	-0.35	-0.35	-0.02*	-0.00*

Table 7: Galaxies, 512 anchors, $k = 2,048$, multi-scale density control: rank-partial Spearman against local R^2 of the shape term, the sphere term and the scalar $\|H^S\|$. * not significant at 0.05.

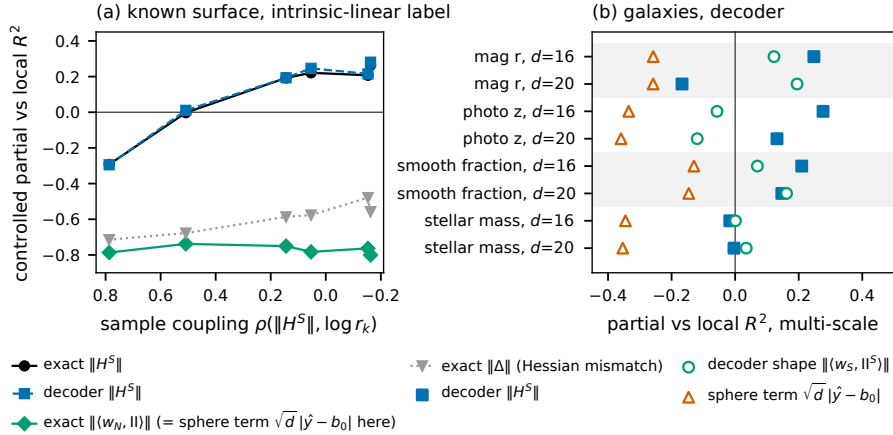


Figure 2: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder's $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).